NATIONAL MATHEMATICS CONTEST

O-LEVEL PAPER 2.

SATURDAY 29TH JUNE, 2019 9:00 AM - 12:15 PM

INSTRUCTIONS TO CANDIDATES AND SUPERVISORS:

(a) *This competition is conducted on the assumption that proper security is maintained. UMS reserves the right, should there be evidence or suspicion of any malpractice, to reject scripts from a particular school/institution.*

(b) *There are two sections and each section carries* 50 *marks. Attempt all questions in exactly three hours fifteen minutes (*$9:00am$ *to* $12:15pm$*).*

(c) *Marks will be awarded for only answers for which a clear and logical layout of the working is provided. A correct solution, poorly presented, will not earn full marks.*

(d) ***Indicate your names, birth date, gender, class, school and district on all your answer sheets.***

(e) *Answer scripts of only qualified students should be returned to the contest coordinator, latest Monday 1st July, 2019.*

(f) *ALL participants who have sat for Paper* 2 *are cordially invited to the certificate and prize giving ceremony on Saturday* 27*th July,* 2019 *at 8:00am at Makerere University, Kampala.*

SECTION A : 5 marks each

Qn. 1. What is the value of x in the equation; $3 \cdot 16^x + 36^x - 2 \cdot 81^x = 0$. [The dot $(\cdot)$ represents multiplication].

Qn. 2. Let $x + \dfrac{1}{x} = 10$. Find the value of $\dfrac{x^3}{970} + \dfrac{1}{970x^3}$.

Qn. 3. Given that $B_1 = 1!, B_2 = 1! + 2!, B_3 = 1! + 2! + 3!, B_4 = 1! + 2! + 3! + 4!, \cdots$. What is the ones digit of the term B_{2019}?

Qn. 4. Given that the function $f(x)$ satisfies $f(1) = 4$ and $f(x) \cdot f(y) = f(x - y) + f(x + y)$. Find $f(5)$.

Qn. 5. In a sequence $\{a_n\}$ for $n \geq 4$ and $n \in \mathbb{N}$ where $\mathbb{N} = \{1, 2, 3, 4, 5, \cdots\}$, the terms are related by $a_n = a_{n-1} + a_{n-3}$. Given that $a_6 = 30$, what is the value of $a_8 - a_1$?

Qn. 6. After John has used 20% of his pocket money and Joy has used 40% of her pocket money, John has twice as much pocket money left as Joy. Find John's original amount of pocket money as a percentage of Joy's original amount of pocket money.

Qn. 7. Suppose a circle is inscribed in a square each of whose sides is one unit in length. Let A_1 be the area of the square minus the area of the circle. Next a second square is inscribed inside the first circle and a second circle is inscribed inside the second square. Let A_2 be the area of the second square minus the area of the second circle. Then a third square is inscribed inside the second circle and a third circle inside the third square. Let A_3 be the area of the third square minus the area of the third circle. If this process is repeated until A_{10} is found, what is $A_1 + A_2 + A_3 + A_4 + A_5$? [HInt: Leave π in your answer.]

Qn. 8. Suppose Juliet can do a job in 4 hours and Janet can do the same job in 5 hours. If they work together for 2 hours and then Juliet quits, how long will it take Janet to finish the job?

Qn. 9. A is a two-digit whole number that does not contain zero as a digit. B is a three-digit whole number, and $A\%$ of B is 400. Find all possible values of A and B.

Qn. 10. Suppose two bicycle riders are headed towards each other, the first is moving at $10mph$ and the second at $15mph$. At the instant the bicyclists are 30 miles apart, a bird flying at $20mph$ and headed in the same direction as the first rider, passes the first rider. When the bird reaches the second rider, it instantaneously changes direction and heads back toward the first rider. When the bird reaches the first rider it again instantaneously changes direction and heads toward the second rider. If the bird continues to go back and forth at $20mph$, how far will it travel from the time the bikers are 30 miles apart until they reach each other?

SECTION B : 10 marks each

Qn. 11. Find the numbers m and n such that $\sqrt{m} - \sqrt{n} = 2\sqrt{2 - \sqrt{3}}$.

Qn. 12. Three men, Peter, John, and James, had to go on a journey of forty kilometres. Peter could walk one kilometre an hour, John could walk two kilometres an hour, and James could ride his bicycle at eight kilometres an hour. They start at the same time. James rode with Peter a certain distance, and dropped him to walk the remainder, rode back to meet John on the way and carried him to their destination, where they all arrived at the same time. How long did the journey take? Assume each person went at a uniform rate throughout and the bicycle could only carry one extra person at a time.

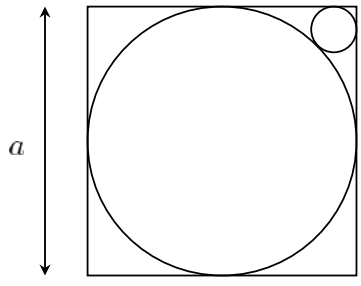
Qn. 13. For $x \notin \left\{0,\ -\sqrt{2},\ \dfrac{-1}{\sqrt{2}}\right\}$. Define $f(x) = \dfrac{-1}{x + \sqrt{2}}$. Using the notation:

$$f^{(1)} = f(x),\ f^{(2)} = f(f(x)),\ f^{(3)} = f(f(f(x)))$$

and in general $f^{(n+1)} = f(f^{(n)}(x))$. Find and simplify $f^{(2019)}(x)$.

Qn. 14. Three gamblers went into a betting shop. At the start of betting they had money in ratios $6 : 5 : 4$ and at the end, their money was in ratios $7 : 6 : 5$. One of the gamblers won $30,000$ UGX. How much money did he have at the end?

Qn. 15. If a circle of radius $\dfrac{a}{2}$ is inscribed in a square of side a, what is the radius of a circle Q that can be inscribed in a corner of the square tangent to two sides of the square and to the circle inscribed in the square?



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SECTION A : 5 marks each

Qn. 1. What is the value of x in the equation; $3 \cdot 16^x + 36^x - 2 \cdot 81^x = 0$. [The dot $(\cdot)$ represents multiplication].

Solution :

$$3{\cdot}16^x{+}36^x{-}2{\cdot}81^x = 0 \Leftrightarrow \frac{3 \cdot 16^x}{36^x}{+}\frac{36^x}{36^x}{-}\frac{2 \cdot 81^x}{36^x} = \frac{0}{36^x} \Leftrightarrow 3{\cdot}\left(\frac{4}{9}\right)^x{+}1{-}2{\cdot}\left(\frac{9}{4}\right)^x = 0.$$

Let $u = \left(\frac{4}{9}\right)^x$ *to have* $3u^2 + u - 2 = 0$ *whose solution set is* $\left\{-1, \frac{2}{3}\right\}$*. Since* $u > 0$ *then* $u = \frac{2}{3}$ *is the only value that satisfies the equation. Therefore* $x = \frac{1}{2}$*.* ■

Qn. 2. Let $x + \frac{1}{x} = 10$. Find the value of $\frac{x^3}{970} + \frac{1}{970x^3}$.

Solution : *Consider* $\left(x + \frac{1}{x}\right)^3 = 10^3$. *On expanding we get*

$$\left(x^3 + \frac{1}{x^3}\right)^3 = 1000 - 3\left(x + \frac{1}{x}\right) = 970.$$

This means that

$$\frac{x^3}{970} + \frac{1}{970x^3} = \frac{1}{970}\left(x^3 + \frac{1}{x^3}\right) = \frac{970}{970} = 1.$$

■

Qn. 3. Given that $B_1 = 1!, B_2 = 1! + 2!, B_3 = 1! + 2! + 3!, B_4 = 1! + 2! + 3! + 4!, \cdots$. What is the ones digit of the term B_{2019}?

Solution :

$$\begin{aligned} B_1 &= 1! = 1, \\ B_2 &= 1! + 2! = 3, \\ B_3 &= 1! + 2! + 3! = 9, \\ B_4 &= 1! + 2! + 3! + 4! = 33, \\ B_5 &= 1! + 2! + 3! + 4! + 5! = 153, \\ B_6 &= 1! + 2! + 3! + 4! + 5! + 6! = 873, \ldots \end{aligned}$$

We note that the ones digit of $n!$ *is zero* $\Rightarrow n \geq 4$ *the ones digit of* B_n *is* 3 *thus the ones digit of* B_{2019} *is* 3. ■

Qn. 4. Given that the function $f(x)$ satisfies $f(1) = 4$ and $f(x) \cdot f(y) = f(x-y) + f(x+y)$. Find $f(5)$.

Solution : *Substitute different values of x in the function relation given,*

$$f(1) \cdot f(0) = f(1+0) + f(1-0) \Leftrightarrow f(0) = 2.$$

For

$$f(1) \cdot f(1) = f(1+1) + f(1-1) \Leftrightarrow f(2) = 14.$$

Also

$$f(2) \cdot f(1) = f(2+1) + f(2-1) \Leftrightarrow f(3) = 52.$$

Now consider,

$$f(3) \cdot f(2) = f(5) + f(1) \Leftrightarrow f(5) = 724.$$

■

Qn. 5. In a sequence $\{a_n\}$ for $n \geq 4$ and $n \in \mathbb{N}$ where $\mathbb{N} = \{1, 2, 3, 4, 5, \cdots\}$, the terms are related by $a_n = a_{n-1} + a_{n-3}$. Given that $a_6 = 30$, what is the value of $a_8 - a_1$?

Solution :

$$\begin{aligned}
a_4 &= a_3 + a_1 \\
a_5 &= a_4 + a_2 = a_2 + a_3 + a_1 \\
a_6 &= a_5 + a_3 = a_2 + 2a_3 + a_1 \\
a_7 &= a_6 + a_4 = a_2 + 3a_3 + 2a_1 \\
a_8 &= a_7 + a_5 = 2(a_2 + 2a_3 + a_1) + 3a_1 \Leftrightarrow a_8 = 2 \times 30 + a_1 \Rightarrow a_8 - a_1 = 60
\end{aligned}$$

■

Qn. 6. After John has used 20% of his pocket money and Joy has used 40% of her pocket money, John has twice as much pocket money left as Joy. Find John's original amount of pocket money as a percentage of Joy's original amount of pocket money.

Solution : *Let x be the original amount of pocket money John had and y be the original amount that Joy had. Given this then*

$$0.8x = 2(0.6y) \Leftrightarrow y = \frac{2}{3}.$$

The required percentage is

$$\left(\frac{x}{y}\right) \times 100\% = \left(\frac{x}{\frac{2x}{3}}\right) \times 100\% = 150\%.$$

■

Qn. 7. Suppose a circle is inscribed in a square each of whose sides is one unit in length. Let A_1 be the area of the square minus the area of the circle. Next a second square is inscribed inside the first circle and a second circle is inscribed inside the second square. Let A_2 be the area of the second square minus the area of the second circle. Then a third square is inscribed inside the second circle and a third circle inside the third square. Let A_3 be the area of the third square minus the area of the third circle. If this process is repeated until A_{10} is found, what is $A_1 + A_2 + A_3 + A_4 + A_5$? [HInt: Leave π in your answer.]

Solution : *The area of the square is* $1 units^2$ *and that of the circle is* $\pi r^2 = \frac{\pi}{4} \Rightarrow A_1 = 1 - \frac{\pi}{4}$. *The side of the second square is* $\frac{1}{\sqrt{2}}$ *and the radius of the second circle is*

$$\frac{1}{2\sqrt{2}} \Rightarrow A_2 = \left(\frac{1}{\sqrt{2}}\right)^2 - \pi\left(\frac{1}{2\sqrt{2}}\right)^2 = \frac{1}{2}\left(1 - \frac{\pi}{4}\right).$$

The diagonal of the third square is twice the radius of the second circle that is $2 \cdot \frac{1}{2\sqrt{2}} = \frac{1}{\sqrt{2}}$ *so the length of each of its sides is* $\frac{1}{2}$ *and the radius of the third circle is*

$$\frac{1}{4} \Rightarrow A_3 = \left(\frac{1}{2}\right)^2 - \pi\left(\frac{1}{4}\right)^2 = \frac{1}{4}\left(1 - \frac{\pi}{4}\right).$$

Continuing in this way we get

$$A_1 + A_2 + A_3 + A_4 + A_5 = \left(1 - \frac{\pi}{4}\right)\left(1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4}\right) = \frac{31}{64}(4 - \pi)$$

■

Qn. 8. Suppose Juliet can do a job in 4 hours and Janet can do the same job in 5 hours. If they work together for 2 hours and then Juliet quits, how long will it take Janet to finish the job?

Solution : *Juliet can do the job in* 4 *hours so her rate is* $\frac{1}{4}$ *of the job per hour. Janet's rate is* $\frac{1}{5}$ *of the job per hour. If they work together then their combined rate is* $\frac{1}{4}+\frac{1}{5}=\frac{9}{20}$. *Using the fact that work done is rate times time we have that,* $\frac{9}{20}\times 2=\frac{9}{10}$ *of the job will be done in the first 2 hours remaining* $\frac{1}{10}$ *of the job. This means that Janet will finish the job in* t *hours thus,* $\frac{1}{5}\times t=\frac{1}{10}\Rightarrow t=0.5$ *of an hour. Thus the work will be done in* $2\frac{1}{2}$ *hours.* ■

Qn. 9. A is a two-digit whole number that does not contain zero as a digit. B is a three-digit whole number, and $A\%$ of B is 400. Find all possible values of A and B.

Solution : $\frac{A}{100}B = 400 \Rightarrow AB = 40000 = 2^6 \times 5^4$. *Since A is a two digit number that does not contain 0 so A can not have both 2 and 5 as a factor. Also $A > 40$ otherwise $B \geq 1000$ which is more than a three digit number thus the only possibility is $A = 2^6$ and $B = 5^4$ thus $A = 64$ and $B = 625$.* ■

Qn. 10. Suppose two bicycle riders are headed towards each other, the first is moving at $10mph$ and the second at $15mph$. At the instant the bicyclists are 30 miles apart, a bird flying at $20mph$ and headed in the same direction as the first rider, passes the first rider. When the bird reaches the second rider, it instantaneously changes direction and heads back toward the first rider. When the bird reaches the first rider it again instantaneously changes direction and heads toward the second rider. If the bird continues to go back and forth at $20mph$, how far will it travel from the time the bikers are 30 miles apart until they reach each other?

Solution : *Since one rider is riding at* $10mph$ *and the other at* $15mph$*, the gap between is being closed at a rate of* $25mph$*. Since* $Time = \dfrac{distance}{speed} = \dfrac{30}{25} = \dfrac{6}{5}$ *hours, so the bird will fly* $20mph$ *for* $\dfrac{6}{5}$ *hours covering a distance* $20 \times \dfrac{6}{5} = 24$ *miles.* ■

SECTION B : 10 marks each

Qn. 11. Find the numbers m and n such that $\sqrt{m} - \sqrt{n} = 2\sqrt{2 - \sqrt{3}}$.

Solution : *Squaring both sides of* $\sqrt{m} - \sqrt{n} = 2\sqrt{2 - \sqrt{3}}$ *gives,*

$$m - 2\sqrt{mn} + n = 8 - 4\sqrt{3}$$

equating similar terms gives $m + n = 8$ *and* $mn = 12$ *whose solution is* $m = 6$ *and* $n = 2$ *since* $m > n$ *as* $2\sqrt{2 - \sqrt{3}} > 0$. ■

Qn. 12. Three men, Peter, John, and James, had to go on a journey of forty kilometres. Peter could walk one kilometre an hour, John could walk two kilometres an hour, and James could ride his bicycle at eight kilometres an hour. They start at the same time. James rode with Peter a certain distance, and dropped him to walk the remainder, rode back to meet John on the way and carried him to their destination, where they all arrived at the same time. How long did the journey take? Assume each person went at a uniform rate throughout and the bicycle could only carry one extra person at a time.

Solution : *Let x be the distance Peter rode with James and y be the distance John walked. Using the formula Time $= \dfrac{distance}{speed}$ on the times taken by the three to arrive, we have,*

Peter: $\dfrac{x}{8}+\dfrac{40-x}{1}$ *where* $\dfrac{x}{8}$ *is when riding with James and* $40-x$ *when walking.*

John: $\dfrac{y}{2}+\dfrac{40-y}{8}$ *where* $\dfrac{y}{2}$ *is when walking and* $\dfrac{40-y}{8}$ *when riding with James.*

James: $\dfrac{x}{8}+\dfrac{x-y}{8}+\dfrac{40-y}{8}$ *where* $\dfrac{x}{8}$ *riding with Peter,* $\dfrac{x-y}{8}$ *when riding back alone and* $\dfrac{40-y}{8}$ *is when riding with John.*

Given that they all arrived at the same time then

$$\frac{x}{8}+\frac{40-x}{1}=\frac{y}{2}+\frac{40-y}{8}\Leftrightarrow 7x+3y=280$$

and

$$\frac{y}{2}+\frac{40-y}{8}=\frac{x}{8}+\frac{x-y}{8}+\frac{40-y}{8}\Leftrightarrow 2x-5y=0.$$

Now solving $7x+3y=280$ *and* $2x-5y=0$ *gives the values of* x *and* y *as* $x=\dfrac{1400}{41}$ *and* $y=\dfrac{560}{41}$ *Substituting back into the equations give the time as* $10\dfrac{5}{41}$ *hours.* ■

Qn. 13. For $x \notin \left\{0,\ -\sqrt{2},\ \dfrac{-1}{\sqrt{2}}\right\}$. Define $f(x) = \dfrac{-1}{x+\sqrt{2}}$. Using the notation:

$$f^{(1)} = f(x),\ f^{(2)} = f(f(x)),\ f^{(3)} = f(f(f(x)))$$

and in general $f^{(n+1)} = f(f^{(n)}(x))$. Find and simplify $f^{(2019)}(x)$.

Solution : *Since* $f(x) = \dfrac{-1}{x+\sqrt{2}}$ *then*

$$f^{(2)}(x) = \frac{-(x+\sqrt{2})}{1+\sqrt{2}x},\quad f^{(3)}(x) = \frac{-(1+\sqrt{2}x)}{x},\quad f^{(4)}(x) = x$$

implying that $f^{(5)}(x) = f(x),\ \ f^{(6)}(x) = f^{(2)}(x)$, *in general* $f^{(n)}(x) = f^{(m)}(x)$ *when* $n \equiv m(mod4)$. *Now*

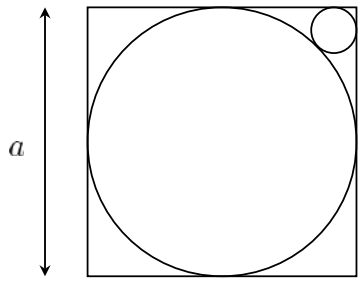
$$2019 \equiv 3(mod4) \Rightarrow f^{(2019)}(x) = f^{(3)}(x) = \frac{-(1+\sqrt{2}x)}{x}.$$

■

Qn. 14. Three gamblers went into a betting shop. At the start of betting they had money in ratios $6:5:4$ and at the end, their money was in ratios $7:6:5$. One of the gamblers won $30,000$ UGX. How much money did he have at the end?

Solution : *Let us assume that the total money involved in the process is* $270x$ *where* 270 *is the product of the sum of the ratios before and after betting. At the start the gamblers had* $108x, 90x$ *and* $72x$*, at the end they have* $105x, 90x$ *and* $75x$*, So the last player is the only one who won and by* $30,000 = 3x$*. The last player therefore had* $75x = 750,000$. ■

Qn. 15. If a circle of radius $\frac{a}{2}$ is inscribed in a square of side a, what is the radius of a circle Q that can be inscribed in a corner of the square tangent to two sides of the square and to the circle inscribed in the square?



Solution : *Let O be the center of the inscribed of radius $\frac{a}{2}$ and P the vertex of the square where the smaller circle is inscribed. Let r be the radius of the smaller circle with center Q. The length of the line segment OP is half the length of a diagonal of the square of side a that is $\frac{\sqrt{2}a}{2}$. The length of OP is also equal to $\frac{a}{2}$ plus r plus the length of the line segment QP. Now construct a square of side $2r$ with P being one of its vertices and circumscribing the smaller circle of radius r. The length of line segment QP is half the length of the diagonal of the square of radius r. Therefore the length of QP is*

$$\sqrt{2}r \Rightarrow \frac{\sqrt{2}a}{2} = \frac{a}{2} + r + \sqrt{2}r.$$

Making r the subject here gives $r = \left(\frac{3-2\sqrt{2}}{2}\right) a.$ ■